

Salon Management System

Software Requirements Specification (SRS)

Baseline specification, produced through a SKED dialogue
(Socratic Knowledge Elicitation & Documentation)

Version 1.0 — Draft baseline, pending SPEC-GATE approval

Field	Value
System name	Salon Management System
Document type	Software Requirements Specification (IEEE 830-informed)
Document status	Baseline, produced via SKED (Lab 2)
Predecessor document	DESCRIPTION.md (project concept note)
Related artifact	2026-09-16-sms-reminder-conflict.md (Lab 1 conflict resolution)
Methodology	Specification-Driven Development (SDD) with SKED elicitation
Platform	Single web application (responsive), single location, single time zone

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1 Introduction

1.1 Purpose

This document specifies the baseline requirements for the Salon Management System: an information system for client record-keeping and service booking for a single-location unisex beauty salon. It supersedes the informal concept note `DESCRIPTION.md` as the controlling requirements artifact once approved at SPEC-GATE (Section 9).

1.2 Scope

The system automates client booking, master schedule management, material stock tracking (general accounting only), and basic financial reporting for one salon location. It does not include a client account system, a loyalty program, automatic payroll calculation, no-show tracking, recipe-based material deduction, or an online payment gateway (Section 2.3).

1.3 Actors

- **Client** — books a service without registration, using name and phone number.
- **Master** — performs services; manages their own schedule and sees their own appointment calendar; authenticates with login/password.
- **Administrator** — manages reference data, appointments, materials, and reporting; single administrator account; authenticates with login/password.

1.4 Document Overview

Section 2 gives the general description. Section 3 is the domain glossary. Section 4 lists functional requirements. Section 5 lists non-functional requirements. Section 7 gives BDD scenarios for key flows. Section 8 is the traceability matrix. Section 9 records the SPEC-GATE decision.

2 General Description

2.1 Superseded Decisions

Two decisions confirmed in `DESCRIPTION.md` §8 are explicitly revised by this SRS, as a result of the SKED dialogue:

1. “*The only SMS notification sent is a reminder before the visit — there is no SMS for booking confirmation or cancellation.*” **Revised:** the system now sends a booking confirmation SMS and a reschedule confirmation SMS (REQ-F-005, REQ-F-009). No additional SMS is sent specifically for cancellation — the client sees the updated status on the confirmation page (REQ-F-004, REQ-F-006).
2. “*The master/administrator do not receive a notification when a new booking is made or cancelled.*” **Revised:** masters and administrators now receive notifications for new, rescheduled, and cancelled appointments, through independently configurable channels (REQ-F-016, REQ-F-023, REQ-F-026).

All other decisions in `DESCRIPTION.md` §8 remain in force and are treated as constraints of this SRS (Section 2.3).

2.2 Product Functions

- Client: multi-service booking, immediate confirmation, free cancellation and reschedule via a link, SMS notifications.
- Master: schedule view, own service/price list, availability management, notifications.
- Administrator: reference data management, appointment management, material tracking, payment recording, reporting, notifications.

2.3 Constraints

- Single salon location, single time zone, no branch network.
- 30-minute appointment slot granularity.
- No client account/password; client identified by name and phone number only.
- No online payment gateway; only the fact of payment is recorded, after service completion.
- No automatic material deduction by service “recipe”; materials tracked as general receipt/issue accounting only.
- No loyalty program, no automatic master payroll calculation, no client no-show tracking.

2.4 Assumptions and Dependencies

- SMS sending is simulated (no real SMS gateway integration in this version), consistent with `DESCRIPTION.md`.
- Expected usage scale is not numerically fixed for this version; no specific throughput or concurrency target is set as an NFR.

3 Domain Glossary

Term	Definition
Client	A person who books a service by providing name and phone number; has no account in the system.
Master	A staff member who performs services; has an individual login account, own service/price list, and own availability schedule.
Administrator	The single staff account with full management access to reference data, appointments, materials, and reports.
Service	A definition of a performable procedure: name, category, base duration. Categories are free-form, created by the administrator (see REQ-F-024); not a fixed enumeration.
MasterService	The link between a master and a service, carrying that master’s individual price for that service.
Appointment	A single client visit: client, master, date/time, status (confirmed automatically on creation), and an unguessable access token used for the confirmation page.
AppointmentService	One of the services included in a single appointment; an appointment may include several.

Term	Definition
Confirmation Page	The web page shown to the client immediately after booking and reachable again via the link sent by SMS; shows appointment details and the cancel/reschedule controls.
Payment	A record of amount, status (paid/unpaid), and payment method for an appointment; created only after the service is completed.
Material	A stock item: name, unit of measure (from the standard list or a custom unit), current stock level.
MaterialTransaction	A recorded receipt or issue of a material: date, quantity.
Notification	An event-driven message (booking confirmation, reschedule confirmation, 24-hour reminder, 2-hour reminder to the client; new / reschedule / cancellation alert to master and administrator), each with a channel and a send status (sent / skipped / failed).
Reminder	The 24-hour or 2-hour client SMS before the appointment; skipped (not sent retroactively) if its scheduled time has already passed at the moment of booking or reschedule.
Notification Preference	A master's or administrator's own on/off setting for a given notification channel (SMS, in-app) and a given event type (new, reschedule, cancellation).

4 Functional Requirements

Each requirement has a stable identifier REQ-F-NNN. Acceptance criteria are stated as observable conditions.

ID	Requirement	Acceptance Criterion
REQ-F-001	The client can select one or more services for a single visit; total duration and cost are the sum of the selected services.	Booking with 2+ services shows a single duration and a single cost equal to the sum of the individual services.
REQ-F-002	The client selects a master, date, and time from that master's available 30-minute slots.	Only slots within the selected master's set availability, and not already booked, are selectable.
REQ-F-003	The appointment is confirmed automatically on creation; no manual approval step exists.	Immediately after booking, the appointment status is "confirmed" with no pending/approval state observable.
REQ-F-004	After a successful booking, the system shows a confirmation page with appointment details and cancel/reschedule links.	The page is displayed right after booking and is reachable again later via the SMS link (REQ-F-005).
REQ-F-005	The system sends the client a booking confirmation SMS (simulated) with a link to the confirmation page.	Exactly one confirmation SMS per successful booking is recorded, containing a valid link to REQ-F-004's page.
REQ-F-006	The client can cancel the appointment via the link, free of charge, with no restriction on notice period.	Cancelling at any time, including minutes before the appointment, succeeds without a fee or error.

ID	Requirement	Acceptance Criterion
REQ-F-007	The client can reschedule the appointment via the link, free of charge, with no restriction on notice period or number of times.	Rescheduling repeatedly, including shortly before the appointment, succeeds each time without a fee or error.
REQ-F-008	On reschedule, the 24-hour and 2-hour reminder times are recalculated from the new appointment time.	After a reschedule, the stored reminder times equal $\text{new_time} - 24\text{h}$ and $\text{new_time} - 2\text{h}$.
REQ-F-009	On reschedule, the system sends a new SMS with the updated appointment details and a link to the (same) confirmation page.	Exactly one reschedule SMS per successful reschedule is recorded.
REQ-F-010	The system sends a 24-hour reminder SMS, unless that time has already passed at the moment of booking or reschedule, in which case it is recorded as skipped.	A booking made $<24\text{h}$ before the appointment shows the 24h reminder with status “skipped”; otherwise “sent” at the correct time.
REQ-F-011	The system sends a 2-hour reminder SMS under the same rule as REQ-F-010.	A booking made $<2\text{h}$ before the appointment shows the 2h reminder with status “skipped”.
REQ-F-012	A master authenticates with login and password; the account is created by the administrator.	A master cannot self-register; login succeeds only for an account created by the administrator.
REQ-F-013	A master can view their own appointment calendar.	The calendar shown to a logged-in master contains only that master’s appointments.
REQ-F-014	A master can view the services they perform and their own prices.	The list shown matches exactly the MasterService records for that master.
REQ-F-015	A master sets their availability using a recurring weekly pattern, specific dates, or both.	Both a weekly-repeating slot and a one-off date-specific slot can be created and take effect for booking.
REQ-F-016	A master receives a notification (SMS and/or in-app, per their own preferences) when one of their own appointments is created, rescheduled, or cancelled.	Disabling all channels for an event type results in no notification of that type being sent to that master.
REQ-F-017	The administrator authenticates with login and password; the system supports a single administrator account.	Only one administrator account exists and can log in.
REQ-F-018	The administrator manages masters, services (with category), MasterService price links, and materials (with unit of measure).	Create / edit / deactivate operations are available for each of these reference entities.
REQ-F-019	The administrator can create, edit, and cancel client appointments.	An administrator-created or edited appointment is reflected identically to a client-made one (subject to REQ-F-002, REQ-F-003).

ID	Requirement	Acceptance Criterion
REQ-F-020	The administrator records material receipt and issue transactions.	Each transaction updates the material's stock level by the recorded quantity.
REQ-F-021	The administrator records payment (amount, status paid/unpaid, method) only after the service has been completed.	Payment cannot be recorded for an appointment whose service has not yet been marked completed.
REQ-F-022	The administrator can view revenue reports for a fixed period (day/week/month) or a custom date range, a popular-services report, and current material stock levels.	Both a preset-period query and a custom from/to date range return a revenue total for that window.
REQ-F-023	The administrator receives a notification (per their own preferences) for every appointment created, rescheduled, or cancelled, across all masters.	An event for any master's appointment produces a notification to the administrator unless disabled.
REQ-F-024	The administrator creates services with a category of their own choosing; the system provides seed example categories (Hair, Coloring, Barbering) but does not restrict categories to a fixed list.	A new service can be saved with a category value not present in the seed examples.
REQ-F-025	The administrator selects a material's unit of measure from the standard list (ml, l, g, kg, pcs) or adds a custom unit.	A material can be saved with a unit outside the standard list.
REQ-F-026	A master or administrator can independently enable or disable each notification channel (SMS, in-app) for each event type (new, reschedule, cancellation).	Toggling one channel/event combination does not change any other channel/event combination's setting.

5 Non-Functional Requirements

ID	Category	Requirement and Acceptance Criterion
REQ-NF-001	Security	Cancellation and reschedule links shall use a non-sequential, non-guessable identifier (e.g., UUID) as the appointment access token. <i>AC</i> : incrementing or guessing a token value does not grant access to another client's appointment.
REQ-NF-002	Usability	The client-facing booking web interface shall be usable on mobile browsers via a responsive layout. <i>AC</i> : all client flows (booking, cancel, reschedule) are completable on a mobile-width viewport without horizontal scrolling of core controls.

ID	Category	Requirement and Acceptance Criterion
REQ-NF-003	Localization	The user interface language shall be English. <i>AC</i> : no interface text, label, or message is displayed in another language.
REQ-NF-004	Localization	Monetary amounts shall be displayed in Ukrainian hryvnia (UAH). <i>AC</i> : all price and payment fields render with a UAH indicator.
REQ-NF-005	Data retention	Appointment and payment history shall be retained indefinitely; no automatic deletion or archiving is implemented. <i>AC</i> : a record created in this version is still retrievable after an arbitrary elapsed time, with no purge job present.
REQ-NF-006	Privacy	Client personal data (name, phone number) shall be handled under standard data-hygiene practice; no additional regulatory compliance mechanism is required for this version. <i>AC</i> : no requirement beyond REQ-NF-001 (unguessable access) governs client data exposure.
REQ-NF-007	Constraint (carried over)	The system shall operate for a single salon location in a single time zone. <i>AC</i> : no location-selection or time-zone-selection control exists in the interface.
REQ-NF-008	Constraint (carried over)	All appointment scheduling shall use 30-minute time slots. <i>AC</i> : no bookable start time falls outside a 30-minute boundary.
REQ-NF-009	Constraint (carried over)	The system shall not integrate with any online payment gateway. <i>AC</i> : no payment-provider redirect or API call exists in the booking or payment flow.

6 External Interface Requirements

6.1 User Interface

Client-facing interface: a responsive website (REQ-NF-002), English (REQ-NF-003), no login. Master and Administrator interfaces: a web dashboard behind login (REQ-F-012, REQ-F-017).

6.2 Communication Interfaces

SMS delivery is simulated for this version (Section 2); no real telecom gateway is integrated. In-app notifications (REQ-F-016, REQ-F-023, REQ-F-026) require no external interface.

7 BDD Scenarios

The following Gherkin-style scenarios cover the primary flows and the significant edge cases identified during the SKED dialogue. Each scenario lists the requirement IDs it exercises.

```
# SCN-01 -- Successful multi-service booking
# Covers: REQ-F-001, REQ-F-002, REQ-F-003, REQ-F-004, REQ-F-005
Feature: Client books multiple services in one visit
  Scenario: Booking two services with a single available master
    Given a master "Olena" offers "Haircut" (30 min) and "Blow-dry" (30 min)
    And "Olena" has an open 60-minute slot on 2026-10-01 at 14:00
    When a client books "Haircut" and "Blow-dry" with "Olena" at that slot
    Then the appointment is confirmed immediately
    And the appointment duration is 60 minutes
```

```
And the appointment cost is the sum of both service prices
And a confirmation page is shown with cancel and reschedule links
And exactly one confirmation SMS is recorded for the client

# SCN-02 -- Free cancellation with no restriction
# Covers: REQ-F-006
Feature: Client cancels an appointment
  Scenario: Cancelling minutes before the appointment
    Given a confirmed appointment starting in 10 minutes
    When the client opens the cancellation link and confirms cancellation
    Then the appointment status becomes "cancelled"
    And no cancellation fee is applied

# SCN-03 -- Free reschedule with reminder recalculation
# Covers: REQ-F-007, REQ-F-008, REQ-F-009
Feature: Client reschedules an appointment
  Scenario: Rescheduling to a new available slot
    Given a confirmed appointment on 2026-10-01 at 14:00
    And a new open slot with the same master on 2026-10-03 at 10:00
    When the client reschedules to the new slot via the link
    Then the appointment time becomes 2026-10-03 10:00
    And the 24-hour reminder is scheduled for 2026-10-02 10:00
    And the 2-hour reminder is scheduled for 2026-10-03 08:00
    And exactly one reschedule SMS is recorded for the client

# SCN-04 -- 24-hour reminder skipped for a late booking
# Covers: REQ-F-010
Feature: Reminder suppression for late bookings
  Scenario: Booking made less than 24 hours in advance
    Given the current time is 2026-10-01 10:00
    When a client books an appointment for 2026-10-02 08:00
    Then the 24-hour reminder is recorded with status "skipped"
    And the 2-hour reminder is scheduled normally for 2026-10-02 06:00

# SCN-05 -- Both reminders skipped for a very late booking
# Covers: REQ-F-010, REQ-F-011
Feature: Reminder suppression for very late bookings
  Scenario: Booking made less than 2 hours in advance
    Given the current time is 2026-10-01 10:00
    When a client books an appointment for 2026-10-01 11:00
    Then the 24-hour reminder is recorded with status "skipped"
    And the 2-hour reminder is recorded with status "skipped"

# SCN-06 -- Master disables SMS but keeps in-app notifications
# Covers: REQ-F-016, REQ-F-026
Feature: Per-channel notification preferences
  Scenario: Master receives only in-app notification for a new booking
    Given master "Olena" has disabled SMS for "new appointment"
    And master "Olena" has enabled in-app for "new appointment"
    When a client books an appointment with "Olena"
    Then "Olena" receives an in-app notification
    And "Olena" receives no SMS notification

# SCN-07 -- Payment recorded only after service completion
# Covers: REQ-F-021
Feature: Payment recording
  Scenario: Administrator attempts to record payment before completion
    Given an appointment that has not been marked as completed
    When the administrator attempts to record a payment for it
```

```

    Then the system rejects the payment recording

Scenario: Administrator records payment after completion
    Given an appointment that has been marked as completed
    When the administrator records a payment of 800 UAH, status "paid", method "cash"
    Then the payment is saved and linked to that appointment

# SCN-08 -- Administrator creates a service with a custom category
# Covers: REQ-F-024
Feature: Free-form service categories
    Scenario: Creating a service outside the seed categories
        Given the seed categories "Hair", "Coloring", "Barbering" exist
        When the administrator creates a service "Eyebrow shaping" with category "
        Cosmetology"
        Then the service is saved with category "Cosmetology"
        And "Cosmetology" becomes selectable for future services

```

8 Traceability Matrix

Columns for Ticket, Test, Code, and Documentation are intentionally left blank at this stage; they are populated in later SDD iterations.

Req. ID	Acceptance Crit.	BDD	Ticket	Test	Code	Doc
REQ-F-001	§4	SCN-01				
REQ-F-002	§4	SCN-01				
REQ-F-003	§4	SCN-01				
REQ-F-004	§4	SCN-01				
REQ-F-005	§4	SCN-01				
REQ-F-006	§4	SCN-02				
REQ-F-007	§4	SCN-03				
REQ-F-008	§4	SCN-03				
REQ-F-009	§4	SCN-03				
REQ-F-010	§4	SCN-04, SCN-05				
REQ-F-011	§4	SCN-05				
REQ-F-012	§4	—				
REQ-F-013	§4	—				
REQ-F-014	§4	—				

Req. ID	Acceptance Crit.	BDD	Ticket	Test	Code	Doc
REQ-F-015	§4	—				
REQ-F-016	§4	SCN-06				
REQ-F-017	§4	—				
REQ-F-018	§4	SCN-08				
REQ-F-019	§4	—				
REQ-F-020	§4	—				
REQ-F-021	§4	SCN-07				
REQ-F-022	§4	—				
REQ-F-023	§4	SCN-06				
REQ-F-024	§4	SCN-08				
REQ-F-025	§4	—				
REQ-F-026	§4	SCN-06				
REQ-NF-001	§5	—				
REQ-NF-002	§5	—				
REQ-NF-003	§5	—				
REQ-NF-004	§5	—				
REQ-NF-005	§5	—				
REQ-NF-006	§5	—				
REQ-NF-007	§5	—				
REQ-NF-008	§5	—				
REQ-NF-009	§5	—				

9 SPEC-GATE Decision

Field	Value
Gate	SPEC-GATE (baseline specification approval)
Artifact under review	This document, <code>salon-management-system-srs-v1.0.tex</code>
Input	<code>DESCRIPTION.md</code> , Lab 1 conflict log, this SKED dialogue
Decision	PENDING — awaiting explicit human approval before commit
Superseded decisions	See Section 2.1
Open items	None identified at baseline; further questions expected in subsequent SDD iterations per project convention.

This gate record is finalized (status changed from PENDING to APPROVED) once the human reviewer confirms the document, per Section 4.3 of `commit-convention.md` conventions for `gate:` artifacts.